

SHRUTI RAVIKUMAR

B.Tech CSE Student

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Profile

Computer Science undergraduate at IIITDM Kancheepuram with hands-on experience in **software development, AI/computer vision, robotics, DevOps, and data analytics**. **Winner of the Bharat Nirman Track at DECODE SIH 2026**, with experience across hackathons, technical projects, Developers Club, and AUV Society.

Achievements

- **Winner — Bharat Nirman Track, DECODE SIH 2026**
- **Google The Big Code 2026** — Advanced to Round 2
- **Vashisht Hackathon 2026** — Participant
- **Bharat Innovation Hackathon** — Participant
- **Adobe Hackathon** — Participant
- **CodeDuelz — Tantra Fiesta'25, IIIT Nagpur** — Participant

Positions of Responsibility

Student Coordinator — The Developers Club, IIITDM Kancheepuram *May 2026 – Present*

- Coordinate technical events, workshops, and developer activities.

Software Subsystem Team Member — AUV Society, IIITDM Kancheepuram *May 2026 – Present*

- Work on robotics software using **ROS1/ROS2, Linux, OpenCV, YOLOv8, and PID control**.
- Explore autonomous navigation, sensors, actuators, and vehicle-control systems.

Projects

Automated Dependency Vulnerability Remediation for GitHub Repositories *DevOps — Software Security*

- Built an automated workflow to identify vulnerable dependencies in GitHub repositories and initiate remediation.
- Worked with dependency management, repository automation, and software supply-chain security.

AURA — AI-Powered Adaptive Traffic Signal System *AI — Computer Vision — Robotics — Optimization*

- Developed the **Vision & Live Data Layer** using YOLOv8 and OpenCV to detect vehicles, pedestrians, cyclists, and emergency vehicles from live camera feeds.
- Estimated vehicle speed, identified pedestrian priority zones and emergency vehicles, and calculated **Time-To-Intersection** for adaptive signal decisions.
- Generated structured, timestamped JSON data for the backend, optimization engine, SUMO digital twin, and live dashboard.

Pitwall Analytics — Interactive Formula 1 Pre-Race Analytics Platform(Not deployed yet) *Python — FastF1 — Data Analytics — Web Development*

- Building an interactive F1 platform focused on **current driver performance at selected circuits**.
- Provides driver analytics, track visualization, historical context, and race insights using automatically updated data.

Dashing Raptor — Ranger Robot *mBot — Robotics*

- Built and programmed a ranger robot using mBot, integrating sensors, movement, and basic autonomous control.

Education

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram *2025 – Present*
B.Tech — Computer Science & Engineering

Abu Dhabi Indian School, Muroor *2011 – 2025*
High School Diploma

Technical Skills

Programming	C, C++, Python
AI & Computer Vision	YOLOv8, OpenCV
Robotics	ROS1, ROS2, Linux, PID Control, TVMC, mBot, Sensors & Actuators, Autonomous Navigation
Web	HTML, CSS
Data	FastF1, MySQL, CSV/File Handling
DevOps	Git, GitHub, Dependency Management, Basic DevOps
Algorithms	Sorting, Recursion, Linked Lists, Graph Algorithms, Problem Solving
Tools	Postman, Thonny, VSCode, TomTom APIs

Certifications

Generative AI with Large Language Models — Coursera

School Leadership & Achievements

Former President, Vice President & Editor — The Outreach Club, Abu Dhabi Indian School

- Led coding activities, workshops, and technology initiatives.

Hackathon Winner — Abu Dhabi University

- **1st Place** — Python Programming
- **2nd Place** — Sustainability in Tech-Product Elevator Pitch